



Grocery Market Basket Analysis

Transforming Shopping Experience at Walmart

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Executive Summary

Business Challenge

Walmart experienced **17% decline in sales** and needed to understand customer purchasing patterns to improve merchandising and increase basket value.

Our Approach

Applied Market Basket Analysis on 38,000+ transactions to discover product associations using the **FP- Growth algorithm**.

Key Outcomes

+24.8%

Basket Size
Increase

\$2K+

Estimated Monthly Revenue
Increase

\$23.5K+

Estimated Annual Revenue
Increase



About Walmart

Founded by **Sam Walton in 1962**, Walmart has grown into the world's largest retail company, recognized for its customer-centric approach and data-driven strategies that continue to shape the global retail landscape.

19

Countries

Global Reach

270M+

Customers

Served Weekly

2.1M+

Associates

Worldwide

Problem Statement

Walmart experienced a 17% decline in sales in late 2015, indicating potential customer satisfaction issues and engagement breakdowns.

Core Question

How can Walmart leverage basket insights to improve product placement, create targeted promotions, and deliver a better shopping experience?

Key Objectives

01

Identify frequently bought-together products

02

Highlight underperforming items

03

Analyze temporal shopping patterns to optimize staffing and promotional timing

04

Uncover strong product associations

05

Recommend bundling and placement strategies

Methodology



Data Generation

Created synthetic grocery transaction data (2014-2015) with over 38,000 rows to replicate Walmart's purchase history



Data Preprocessing

Cleaned dataset, removed duplicates, standardized product names, and created transactional baskets



Exploratory Analysis

Analyzed purchase patterns, identified top products, and examined seasonal trends



Association Mining

Applied the FP-Growth algorithm to extract high-confidence product combinations using support (≥ 0.002), confidence (≥ 0.15), and lift (> 3)

Dataset Description

38K+

Total Transactions

Grocery purchases analyzed
across two years (2014-2015)

167

Products

Diverse product categories from fresh
products to household items

2

Years

Complete dataset spanning
2014-2015 shopping periods

Exploratory Data Analysis

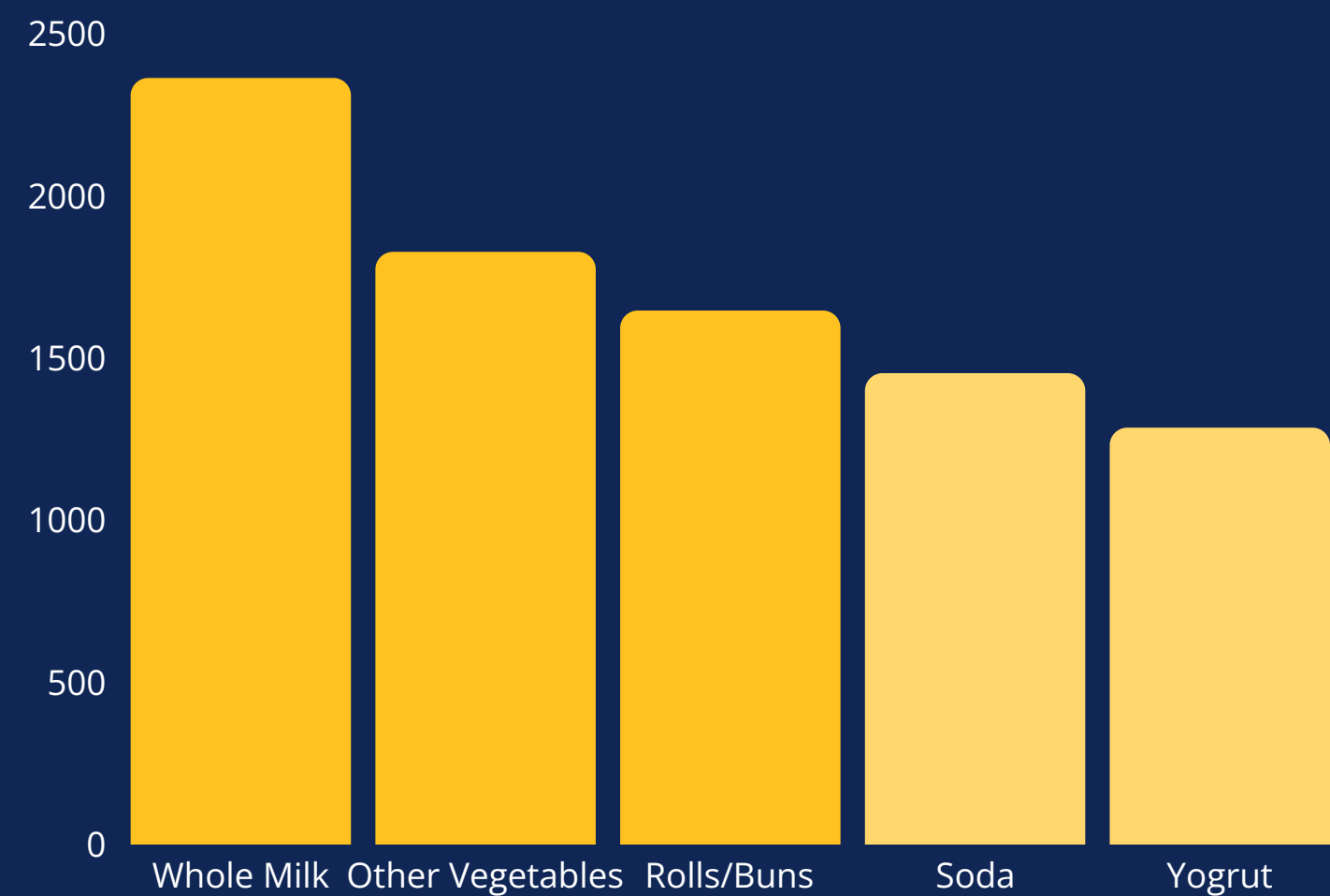
Understanding Purchase Patterns & Customer Behavior



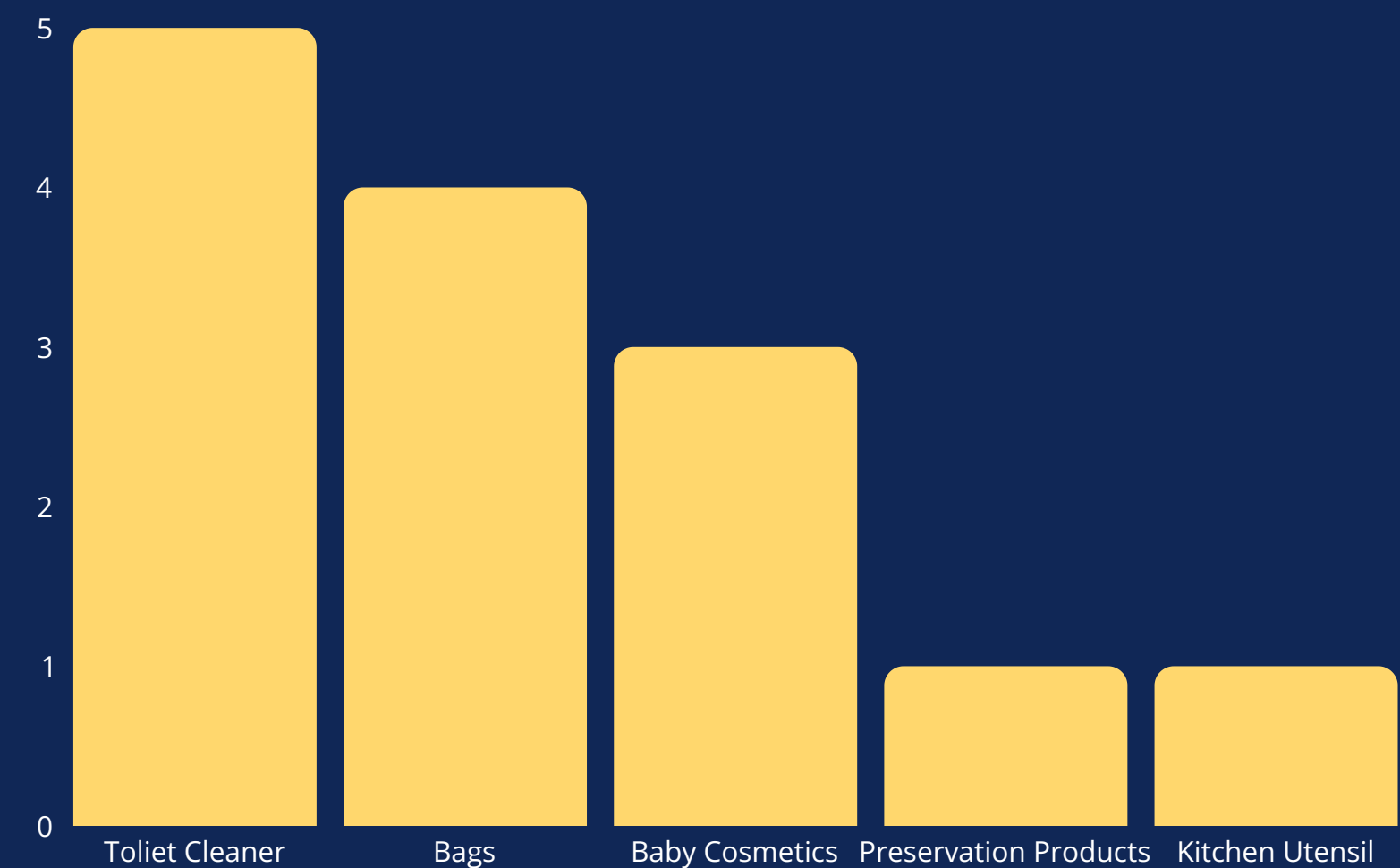
Essential Staples Dominate Grocery Baskets

Core staples like Milk, Vegetables, and Bread lead sales, contributing over 40% of total transactions, while categories like Makeup Remover and Kitchen Utensils lag.

Top 5 Products by Sales Volume

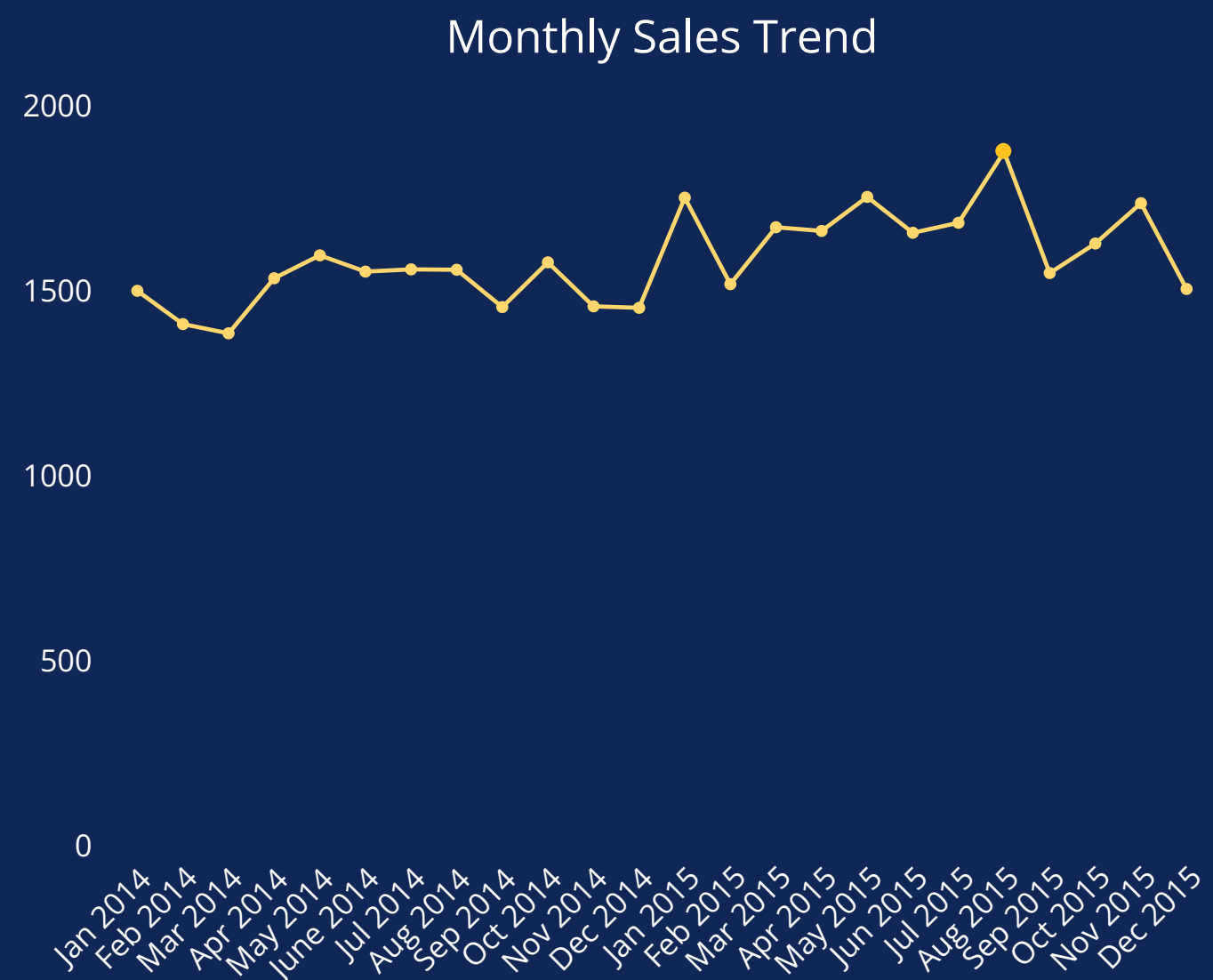
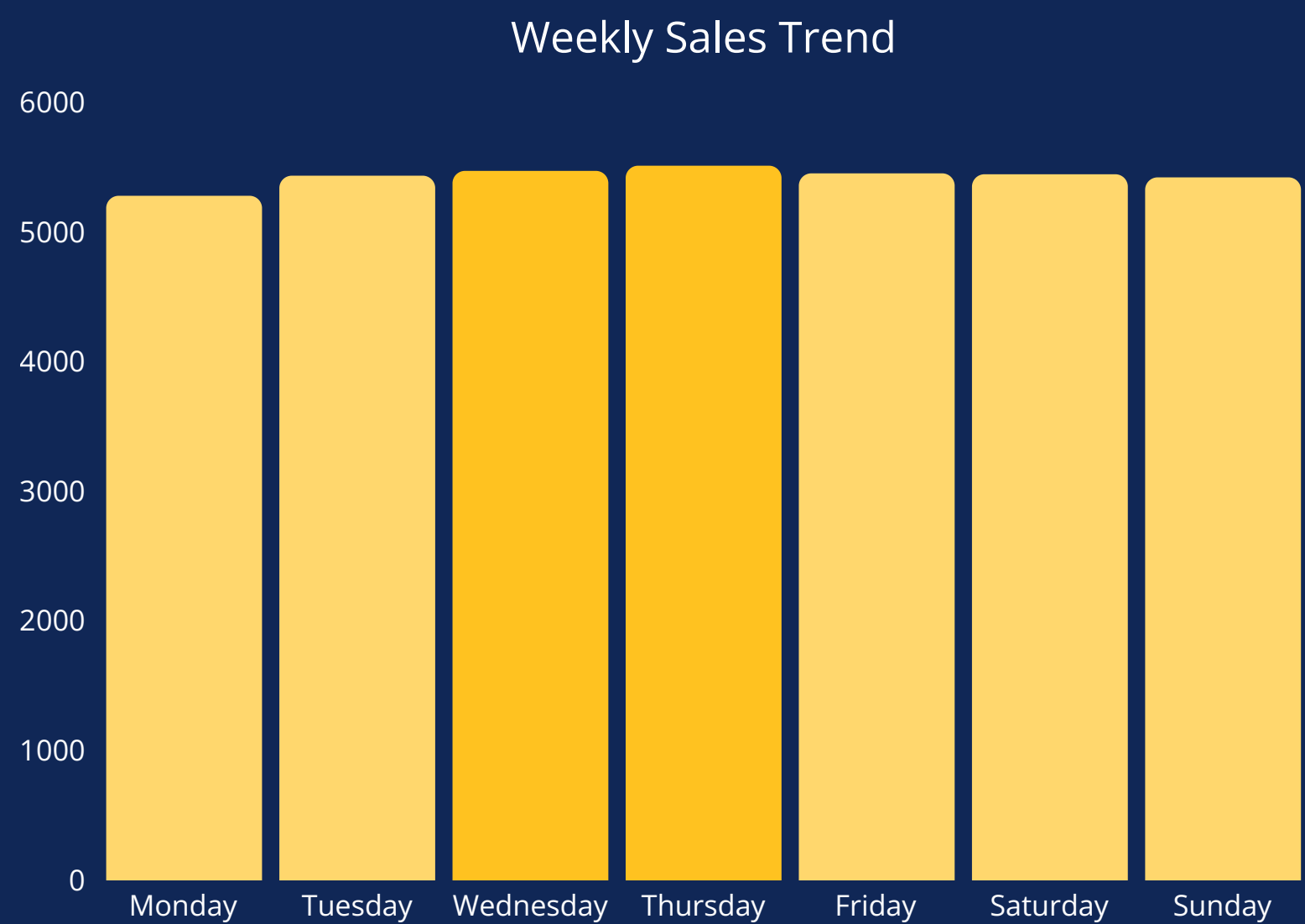


Bottom 5 Products by Sales Volume



Midweek Peaks and August Spikes Highlight Key Promotional Windows

Wednesdays and Thursdays show the highest shopping frequency, while August sees seasonal demand surges.



Market Basket Analysis

Discovering Product Associations & Purchase Patterns



Product Associations

These strong product associations provide valuable insights for strategic product placement and promotional bundling, enhancing the overall shopping experience.



Potato Products + Beef

45% confidence
3.8x Lift

Strong meal pairing



Kitchen Towels + UHT Milk

30% confidence
3.8x Lift

Household essentials



Flour + Mayonnaise

23% confidence
3.3x Lift

Baking combination

Notes:

Each product pair demonstrates a lift above 3, validating strong and statistically significant purchase relationships.



Strategic Recommendations



Product Bundling

Create combo deals for high-association pairs like Potato Products & Beef or Flour & Mayonnaise to increase average basket value and drive cross-selling.



Strategic Placement

Rearrange shelf layouts to place co-purchased items near each other, driving impulse purchases and improving customer convenience.



Targeted Promotions

Deploy time-based campaigns during peak shopping days (Wednesday-Thursday) and promote underperforming items alongside popular products.

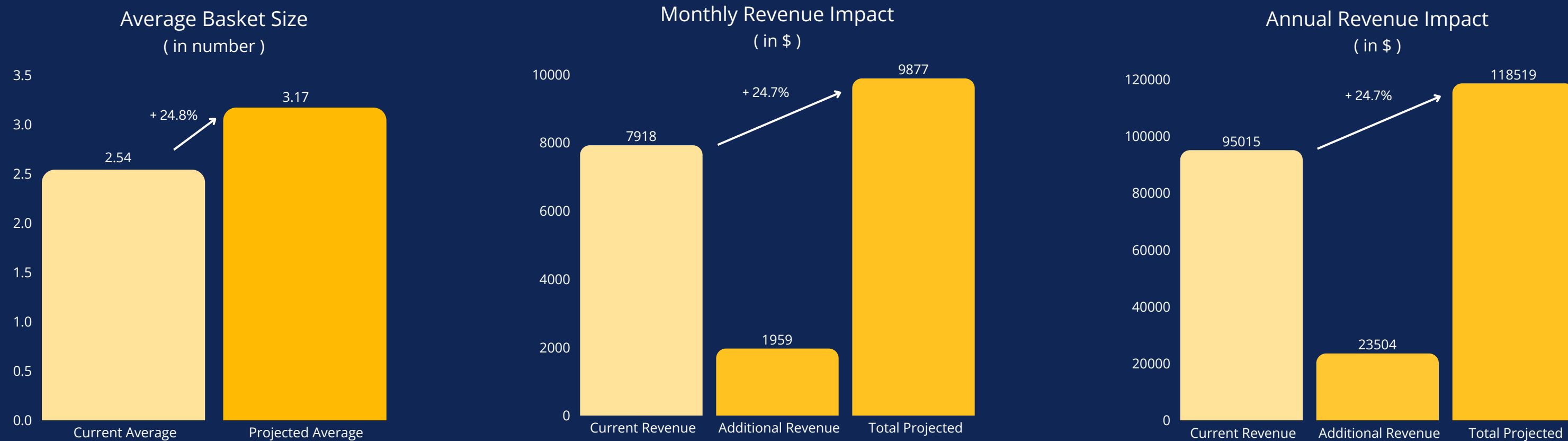


Personalized Offers

Leverage digital channels and receipt-based recommendations to deliver personalized offers based on individual co-purchase behavior patterns.

Projected Buisness Impact

Wednesdays and Thursdays show the highest shopping frequency, while August sees seasonal demand



Assumptions:

- Average item price: \$5.00
- Bundling adoption rate: 10% of transactions
- Average lift from top 5 rules: 3.27x
- Conservative estimate - actual impact may be higher with effective implementation

Thank You

For any queries related to this project, feel free to connect me on the [Linkedin](#).